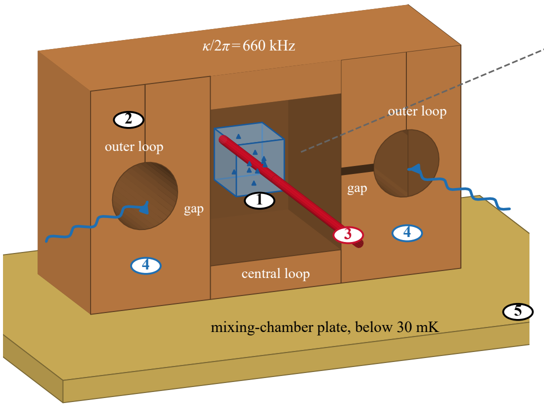
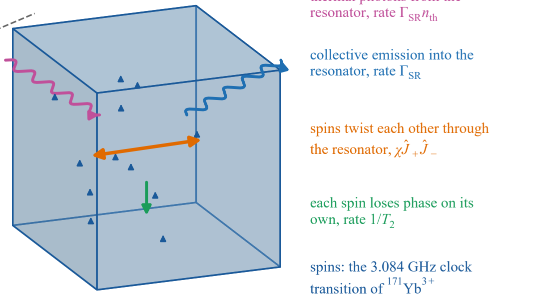
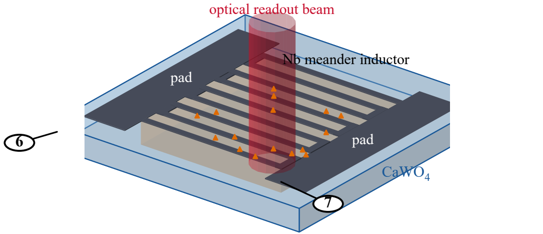
(a) the demonstrated experiment: crystal in a loop-gap microwave resonator



(b) what the model follows inside the crystal



(c) proposed planar superconducting resonator on the same crystal (not built)



Key

1. $^{171}Yb^{3+}:CaWO_4$ crystal, 4.4 × 4.6 × 5 mm, 4.96 ppm ^{171}Yb, glued in the central loop
2. metal loop-gap resonator: central loop, two outer loops, narrow gaps ($g/2\pi = 15$ mHz, $V_m = 275$ mm^3)
3. 973 nm light through the crystal: optical pumping into $|\downarrow\rangle$ and absorption readout of the populations
4. microwave antennas at the outer loops: spin control pulses and transmission measurement
5. mixing-chamber plate of the dilution refrigerator (spin temperature 80 mK, $n_{th} = 0.19$)
6. $CaWO_4$ chip with a niobium film patterned into a meander inductor between two capacitor pads
7. read-out volume under the inductor, where the microwave field is nearly uniform (coupling spread D)